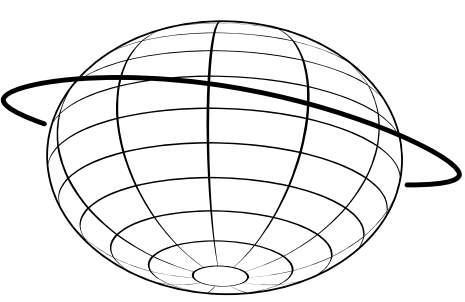
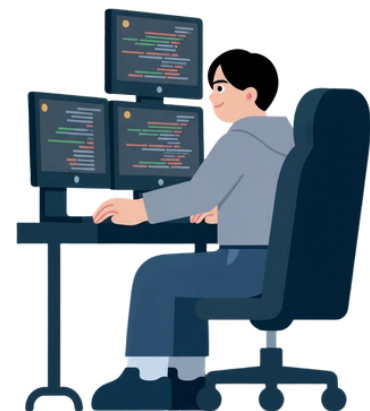


- Problem Statement Title - [AI-Powered Railway Intelligence Platform for Safety, Predictive Maintenance, Smart Rake Allocation, and Real-Time Operations Monitoring.](#)
- Theme - [Railways](#)
- Team Name - [Innovexa](#)
- [Github Link](#)



RAILWAY INTELLIGENCE PLATFORM

PROBLEMS

1

Inefficient Rake Utilization

- Empty rakes travel long distances without cargo.
- Poor allocation reduces fleet utilization.
- High operational and fuel costs.
- Freight revenue opportunities are lost.

2

Track Deterioration & Safety Risks

- Track defects are often detected late.
- Manual inspections are slow and costly.
- High-risk segments remain unnoticed.
- Delayed maintenance increases risk.

3

Rolling Stock Failures

- Wheel and bearing failures cause breakdowns.
- Unexpected failures disrupt operations.
- Maintenance is mostly reactive.
- Increased downtime and repair costs.

4

Forward Safety Challenges

- Obstacles on tracks cause accidents.
- Human visibility is limited at night.
- Animals and vehicles remain undetected.
- Delayed response increases collision risk.

5

Fragmented Intelligence

- Data is scattered across multiple systems.
- No unified monitoring platform exists.
- Limited real-time operational awareness.
- Decision-making remains largely manual.

RAILGUARD

RailGuard AI is an AI-powered railway intelligence platform that integrates Machine Learning, Computer Vision, IoT sensors, and Digital Twin technology to improve railway safety, optimize rake allocation, predict failures, monitor track health, and enable real-time operational decision making. The system reduces operational costs, increases freight revenue, and enhances overall network efficiency.

CORE FEATURES

1

YOLO-Based Forward Vision System

- Detects obstacles, vehicles, animals, and unauthorized persons on railway tracks in real time.
- Generates instant alerts for operators to improve safety and collision prevention.

2

AI-Powered Rake Reallocation

- App and dashboards in 10+ languages, with voice/text emergency access.
- Makes the system accessible to all tourists, including elderly and differently-abled.

3

Digital Twin Command Center

- Provides a live network-wide operational view of rakes, tracks, alerts, and decisions.
- Supports data-driven decision-making through real-time analytics and AI insights.

4

Multilingual Support System

- Supports multiple regional languages for alerts and dashboards.
- Improves accessibility and communication across railway operations.

5

Predictive Asset Health Monitoring

- Monitors track defects, wheel bearings, and axle vibrations using AI.
- Predicts failures early to improve safety and reduce downtime.



TECHNICAL APPROACH – RAILGUARD AI

DATA COLLECTION LAYER

IoT SENSOR LAYER

- Vibration Sensors
- Temperature Sensors
- Sound / Noise Sensors
- Wheel Bearing Sensors
- Axle Load Sensors
- Track Geometry Sensors



RAILWAY OPERATIONS DATA

- Cargo & Booking Data
- Route & Schedule Data
- Fleet / Rake Data
- Maintenance History
- Station & Yard Data
- Historical Performance



CAMERA & VISION DATA

- Track Cameras
- Forward Vision Streams
- Obstacle Images
- Weather Data
- Night Vision Feeds



AI / ML PROCESSING LAYER

PREDICTIVE MAINTENANCE ENGINE



- Monitors Track & Wheel Bearing Health
- Vibration, Temp, Noise
- Anomaly Detection
- RUL Prediction



Tech: Scikit-learn

TRACK HEALTH PREDICTION



- Detects Track Defects
- Risk Scoring
- Line Health Index
- Priority Ranking



Tech: Python, Pandas

YOLOv11 OBSTACLE DETECTION



- Real-time Obstacle Detection
- Animals / Vehicles / Humans / Objects
- Early Warning Alerts



YOLO YOLOv11

XGBOOST REVENUE PREDICTION



- Demand Forecasting
- Revenue Estimation
- Empty Rake Reduction
- Profit Maximization



Tech: XGBoost

AI RAKE REALLOCATION ENGINE



- Rake Availability
- Route Suitability
- Demand & Revenue Optimization
- Best Reallocation Plan



Tech: Scikit-learn

RISK ASSESSMENT ENGINE



- Multi-factor Risk Score
- Failure Probability
- Safety Risk Alerts
- Mitigation Suggestions



Tech: Python

DIGITAL TWIN & DECISION CORE

LIVE NETWORK DIGITAL TWIN



FLEET HEALTH MONITOR



EVENT SIMULATION



- What-if Analysis
- Demand Simulation
- Disruption Impact

PREDICTIVE ANALYTICS



- Trend Analysis
- Failure Prediction
- Performance Insights

ALERT MANAGEMENT



- Real-time Alerts
- Risk Notifications
- Escalation Rules

AI DECISION ENGINE



Aggregates Insights from all Models and Generates Optimal Decisions

BACKEND & DATA LAYER

API GATEWAY (FASTAPI)



AUTHENTICATION & AUTHORIZATION



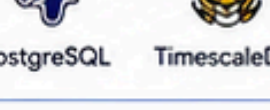
MICROSERVICES ARCHITECTURE



DATA PROCESSING PIPELINE



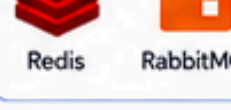
DATABASE LAYER



PostgreSQL

TimescaleDB

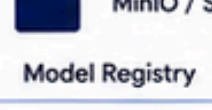
CACHE & QUEUE



Redis

RabbitMQ

FILE & MODEL STORE



MinIO / S3

Model Registry

OUTPUT & ACTION LAYER

AI REALLOCATION DECISIONS

Optimized rake allocation for maximum revenue & minimum empty run

MAINTENANCE ALERTS

Proactive maintenance alerts to reduce failures & downtime

TRACK RISK WARNINGS

High-risk track segments with priority for action

OBSTACLE DETECTION ALERTS

Real-time forward vision alerts for safety

REVENUE OPTIMIZATION INSIGHTS

Revenue forecasts & actionable business recommendations

SUSTAINABILITY REPORTS

Fuel saved, Emission reduced, Green impact reports

REAL-TIME NOTIFICATIONS

Web, Mobile & Email Notifications for critical updates

USER INTERFACE LAYER



NETWORK MAP DASHBOARD

Live Train Movement, Rake Position, Route Conditions



FLEET TRIAGE DASHBOARD

Fleet Health, Alerts, Maintenance Status



REALLOCATION DASHBOARD

AI Recommended Reallocation Plans & Insights



TRACK HEALTH MONITOR

Track Condition, Defect Alerts, Risk Scores



FORWARD VISION SYSTEM

Real-time Obstacle Detection & Alerts



SUSTAINABILITY DASHBOARD

Emission Saved, Fuel Saved, Green Impact



MULTILINGUAL SUPPORT

Dashboard & Alerts in Multiple Languages

TECHNOLOGIES USED



CORE AI MODELS

- 1 AI Rake Reallocation Engine – Maximizes utilization & revenue
- 2 Predictive Asset Health Monitoring – Early failure detection
- 3 Track Health Risk Predictor – Identifies high-risk track segments
- 4 YOLOv11 Forward Vision System – Real-time obstacle detection
- 5 Digital Twin Simulation Engine – Real-time network intelligence
- 6 Revenue Prediction Model – Demand & revenue forecasting



FEASIBILITY AND VIABILITY

Operational Feasibility

- Integrates with existing railway monitoring systems, IoT devices, and operational databases.
- Supports phased deployment across stations, yards, freight corridors, and railway zones.
- Designed for continuous operation with low-latency decision support.

Integrated Workflow

Data Collection

IoT sensors, cameras, railway databases

AI Processing

ML models analyze track conditions, rake health and obstacle detection.

Risk Assessment

System calculates risk scores and identifies maintenance.

Decision Engine

AI recommends optimal rake allocation and maintenance actions.

Alert Generation

Critical alerts are sent to operators, maintenance teams, and control centers.

Continuous Monitoring

Digital Twin dashboard updates live network conditions and asset status.

Technical Feasibility

- Built using Python, FastAPI, React, and modern ML frameworks.
- Supports real-time IoT data ingestion and analytics.
- Digital Twin architecture enables network-wide visualization and simulation.
- Scalable microservice design supports deployment across multiple railway zones.
- Compatible with cloud and on-premise railway infrastructure.

Security & Data Privacy

- Role-based access control for railway personnel.
- Secure API communication using encrypted protocols.
- Sensor and operational data protected through authentication mechanisms.
- Audit logs maintain accountability for critical decisions and alerts.

Regulatory & Institutional Viability

- Aligns with Indian Railways digital transformation initiatives.
- Supports predictive maintenance and railway safety objectives.
- Enhances operational efficiency without major infrastructure changes.
- Can be extended across freight, passenger, and high-speed rail networks.

Economic Viability

Category	Type	Estimated Cost (INR)	Cost-Cutting Measures
Software Development	Non-Recurring	₹0 – ₹50,000	Student-built MVP using open-source tools
Cloud Infrastructure	Non-Recurring	₹20,000 – ₹40,000 / yr	Free-tier AWS/GCP
IoT Sensor Deployment	Recurring	₹50,000 – ₹1,50,000	Deployment on selected assets
AI Model Training	Recurring	₹0 – ₹20,000	Open-source datasets & local GPU training
Maintenance & Support	Recurring	₹10,000 – ₹30,000 / yr	Automated monitoring

Total Estimated One Year Cost

Expense Type	Approx. Total
One-Time Development	₹3–7 Lakhs
Annual Operational Cost	₹50,000–₹1.3 Lakhs
➡ Total Year 1 MVP Cost	₹3.5–8 Lakhs

IMPACT AND BENIFITS



Social Benefits

- Enhances railway safety through early detection of track defects and obstacles.
- Reduces accident risks for passengers, staff, and railway assets.
- Supports safer and more reliable freight transportation across the network.
- Enables data-driven decision making for railway authorities.



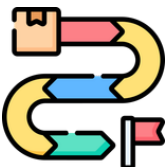
Economic Benefits

- Reduces empty rake movement, improving asset utilization.
- Lowers maintenance costs through predictive maintenance.
- Increases freight revenue through AI-powered rake allocation.
- Minimizes operational losses caused by failures and delays.
- Reduces fuel consumption and carbon emissions.



Operational Benefits

- Real-time monitoring of railway assets and infrastructure.
- Automated risk assessment and alert generation.
- Faster maintenance planning and resource allocation.
- Improved network visibility through Digital Twin technology.
- Better fleet utilization and operational efficiency.



Future Roadmap

- Pilot deployment on selected freight corridors.
- Integration with live railway IoT sensors and control centers.
- Expansion to passenger rail operations.
- Advanced Digital Twin simulations for nationwide railway networks.
- AI-assisted autonomous maintenance planning and scheduling.

COMPARITIVE PERSPECTIVE

Feature	Existing Rule-Based / Isolated ML Schemes	Proposed Context-Aware TrustGraph System
Track Monitoring	Manual inspections	AI-powered real-time monitoring
Rake Allocation	Manual planning	Intelligent AI-based optimization
Obstacle Detection	Limited surveillance	YOLOv11 real-time detection
Maintenance	Reactive maintenance	Predictive maintenance alerts
Network Visibility	Fragmented systems	Unified Digital Twin dashboard